

MCQs from: عشر درجات قبل الميـد.pdf (pdf1_arabic.txt)

Instructions: Choose the best answer for each question.

1. What is another name for First-Order Logic (FOL)? A) Propositional Logic B) Boolean Logic C) Predicate Logic D) Fuzzy Logic **Correct Answer: C**
2. Which of the following is NOT a basic element of First-Order Logic as listed in the document? A) Constants B) Variables C) Algorithms D) Predicates **Correct Answer: C**
3. In the FOL sentence `tall(John)`, what does `tall` represent? A) Variable B) Constant C) Function D) Predicate **Correct Answer: D**
4. In the FOL sentence `friends(Arun, Vijay)`, what do `Arun` and `Vijay` represent? A) Predicates B) Variables C) Constants (representing objects/subjects) D) Functions **Correct Answer: C**
5. Which quantifier means "for all" or "true for all"? A) $\exists$ (Existential) B) $\forall$ (Universal) C) $\neg$ (Negation) D) $\rightarrow$ (Implication) **Correct Answer: B**
6. Which quantifier means "there exists" or "true for at least one"? A) $\forall$ (Universal) B) $\exists$ (Existential) C) $\wedge$ (Conjunction) D) $\vee$ (Disjunction) **Correct Answer: B**
7. What connective is typically used with the Universal Quantifier ($\forall$) when translating sentences? A) $\wedge$ (Conjunction / AND) B) $\vee$ (Disjunction / OR) C) $\rightarrow$ (Implication / IF...THEN) D) $\neg$ (Negation / NOT) **Correct Answer: C**
8. What connective is typically used with the Existential Quantifier ($\exists$) when translating sentences? A) $\rightarrow$ (Implication / IF...THEN) B) $\wedge$ (Conjunction / AND) C) $\vee$ (Disjunction / OR) D) $=$ (Equality) **Correct Answer: B**
9. How would you represent the sentence "Every cat drinks milk" in FOL? A) $\exists x: \text{cat}(x) \wedge \text{drinks}(x, \text{milk})$ B) $\forall x: \text{cat}(x) \rightarrow \text{drinks}(x, \text{milk})$ C) $\forall x: \text{cat}(x) \wedge \text{drinks}(x, \text{milk})$ D) $\exists x: \text{cat}(x) \rightarrow \text{drinks}(x, \text{milk})$ **Correct Answer: B**
10. How would you represent the sentence "Someone is laughing" in FOL? A) $\forall x: \text{laughing}(x)$ B) $\neg \exists x: \text{laughing}(x)$ C) $\exists x: \text{laughing}(x)$ D) `laughing(Someone)` **Correct Answer: C**

11. According to the properties of quantifiers, is $\forall x \forall y$ the same as $\forall y \forall x$? A) Yes B) No C) Only if x and y are the same D) Depends on the predicate **Correct Answer: A**
12. According to the properties of quantifiers, is $\exists x \forall y$ the same as $\forall y \exists x$? A) Yes B) No C) Only if the domain is finite D) Depends on the predicate **Correct Answer: B**
13. What does the FOL sentence $\exists x \forall y \text{ loves}(x, y)$ mean? A) Everyone loves everyone. B) Everyone is loved by at least one person. C) There is a person who loves everyone. D) There is a person who is loved by everyone. **Correct Answer: C**
14. What does the FOL sentence $\forall y \exists x \text{ loves}(x, y)$ mean? A) Everyone loves everyone. B) Everyone is loved by at least one person. C) There is a person who loves everyone. D) Someone loves someone. **Correct Answer: B**
15. Which statement demonstrates quantifier duality? A) $\forall x P(x)$ is equivalent to $\neg \exists x \neg P(x)$ B) $\forall x P(x)$ is equivalent to $\exists x P(x)$ C) $\exists x P(x)$ is equivalent to $\forall x \neg P(x)$ D) $\forall x P(x)$ is equivalent to $\neg \forall x P(x)$ **Correct Answer: A**
16. How is "Nobody loves George" represented in FOL according to the document? A) $\forall x: \text{loves}(x, \text{George})$ B) $\exists x: \text{loves}(x, \text{George})$ C) $\neg \exists x: \text{loves}(x, \text{George})$ D) $\neg \forall x: \text{loves}(x, \text{George})$ **Correct Answer: C**
17. How is "All men are mortal" represented in FOL? A) $\forall x: \text{men}(x) \wedge \text{mortal}(x)$ B) $\exists x: \text{men}(x) \rightarrow \text{mortal}(x)$ C) $\forall x: \text{men}(x) \rightarrow \text{mortal}(x)$ D) $\exists x: \text{men}(x) \wedge \text{mortal}(x)$ **Correct Answer: C**
18. In the context of expert systems, what is the role of the Inference Engine? A) Stores facts and rules. B) Provides the user interface. C) Processes queries using rules and facts. D) Explains the reasoning process. **Correct Answer: C**
19. What is the Knowledge Base in an expert system? A) The component that interacts with the user. B) The component that stores rules and facts. C) The component that performs reasoning. D) The component that stores case-specific facts. **Correct Answer: B**
20. What is Forward Chaining also known as? A) Goal-directed inference B) Hypothesis-driven inference C) Data-directed inference D) Rule-based inference **Correct Answer: C**
21. How does Forward Chaining work? A) Starts with a goal and works backward to find supporting facts. B) Starts with known facts and applies rules to derive new facts. C) Asks the user questions to determine the goal. D) Uses probabilities to infer conclusions. **Correct Answer: B**

22. In forward chaining, what triggers the firing of a rule? A) The system reaching a specific goal. B) Data in the working memory matching the rule's conditions (IF clause). C) The user asking a specific question. D) The absence of data in the working memory. **Correct Answer: B**
23. What is Backward Chaining also known as? A) Data-directed inference B) Fact-driven inference C) Goal-directed inference D) Memory-driven inference **Correct Answer: C**
24. How does Backward Chaining work? A) Starts with known facts and derives conclusions. B) Starts with a hypothesis (goal) and tries to prove it by finding rules that conclude it. C) Randomly applies rules until a solution is found. D) Matches data patterns in the working memory. **Correct Answer: B**
25. In backward chaining, what does the system search for initially? A) Facts that match rule conditions. B) Rules whose conclusions (THEN clause) match the current goal. C) All possible rules in the knowledge base. D) User input to guide the search. **Correct Answer: B**
26. In the forward chaining medical example, what was the first assertion added to working memory based on the initial facts? A) fever B) itchiness C) nasal congestion D) virosis **Correct Answer: C** (Derived from 'runny nose' using R2)
27. In the forward chaining medical example, which rule firing led directly to the final diagnosis of 'influenza'? A) R2 B) R6 C) R1 D) R4 **Correct Answer: C** (Required 'nasal congestion' and 'virosis')
28. In the backward chaining medical example, what was the initial goal? A) diagnose(fever) B) diagnose(virosis) C) diagnose(influenza) D) assert(nasal congestion) **Correct Answer: C**
29. In the backward chaining example, when trying to prove 'virosis' (goal from R1), which rule was fired first? A) R2 B) R4 C) R5 D) R6 **Correct Answer: D** (R6 concludes 'virosis')
30. What is the purpose of the Universal Quantifier ($\forall$)? A) To state that a property holds for at least one object. B) To state that a property holds for all objects in the domain. C) To negate a logical statement. D) To connect two logical statements. **Correct Answer: B**
31. What is the purpose of the Existential Quantifier ($\exists$)? A) To state that a property holds for all objects in the domain. B) To state that a property holds for at least one object in the domain. C) To represent equality between objects. D) To represent functions. **Correct Answer: B**

32. Which element of FOL represents specific objects in the world, like 'Pune' or 'Ajay'?
A) Variables B) Predicates C) Constants D) Functions **Correct Answer: C**
33. Which element of FOL represents properties or relations, like 'Brother' or 'Father'?
A) Variables B) Predicates C) Constants D) Functions **Correct Answer: B**
34. Which element of FOL acts as a placeholder, like 'x' or 'y'? A) Variables B) Predicates
C) Constants D) Functions **Correct Answer: A**
35. Consider the statement: "When you paint a block with green paint, it becomes green." Why is FOL better suited than propositional logic for this? A) Propositional logic cannot handle colors. B) FOL can make general statements about all blocks using quantifiers. C) Propositional logic does not have implication. D) FOL uses predicates, which are more expressive. **Correct Answer: B**
36. In the backward chaining example, why did the goal 'body aches' (from R3) fail? A) The rule R3 was incorrect. B) 'body aches' was already proven false. C) There was no information in working memory or rules to support it. D) The system switched to forward chaining. **Correct Answer: C**
37. What is the role of 'Working Memory' in the context of forward chaining described?
A) Stores the permanent rules. B) Stores the initial goal. C) Stores environmental inputs, data, and derived facts. D) Stores the explanation for the conclusion.
Correct Answer: C
38. If a forward chaining system has the fact 'A' and the rule 'IF A THEN B', what happens? A) The system asks the user if 'B' is true. B) The system adds 'B' to the working memory. C) The system searches for rules concluding 'A'. D) The system stops. **Correct Answer: B**
39. If a backward chaining system has the goal 'C' and the rule 'IF B THEN C', what is the next step? A) Add 'C' to the working memory. B) Set 'B' as a new subgoal. C) Ask the user if 'C' is true. D) Find facts that match 'B'. **Correct Answer: B**
40. Translate "Mary loves everyone" assuming the domain D contains only humans. A) $\exists x \text{ love}(\text{Mary}, x)$ B) $\forall x \text{ love}(\text{Mary}, x)$ C) $\forall x (\text{person}(x) \rightarrow \text{love}(\text{Mary}, x))$ D) $\text{love}(\text{Mary}, \text{Everyone})$ **Correct Answer: B**
41. Translate "Mary loves everyone" assuming the domain D contains humans and non-humans. A) $\exists x \text{ love}(\text{Mary}, x)$ B) $\forall x \text{ love}(\text{Mary}, x)$ C) $\forall x (\text{person}(x) \rightarrow \text{love}(\text{Mary}, x))$ D) $\forall x (\text{person}(x) \wedge \text{love}(\text{Mary}, x))$ **Correct Answer: C**
42. Why is $\forall x (\text{person}(x) \wedge \text{love}(\text{Mary}, x))$ considered incorrect for "Mary loves everyone" if the domain includes non-humans? A) It uses the wrong

quantifier. B) It implies everything in the universe is a person. C) It uses the wrong connective. D) It doesn't specify who Mary is. **Correct Answer: B**

43. How can "No one talks" be represented in FOL? A) $\exists x \text{ talk}(x)$ B) $\forall x \text{ talk}(x)$ C) $\neg \exists x \text{ talk}(x)$ D) $\text{talk}(\text{NoOne})$ **Correct Answer: C** (Equivalent to $\forall x \neg \text{talk}(x)$)

44. How is "Everyone loves himself" represented in FOL? A) $\forall x \exists y \text{ love}(x, y)$ B) $\exists x \text{ love}(x, x)$ C) $\forall x \text{ love}(x, x)$ D) $\forall x \text{ love}(x, \text{Everyone})$ **Correct Answer: C**

45. How is "Everyone loves everyone" represented in FOL? A) $\forall x \text{ love}(x, x)$ B) $\exists x \exists y \text{ love}(x, y)$ C) $\forall x \exists y \text{ love}(x, y)$ D) $\forall x \forall y \text{ love}(x, y)$ **Correct Answer: D**

46. What does the FOL representation $\forall x ((\text{student}(x) \ \& \ x \neq \text{George}) \rightarrow \text{smile}(x))$ mean? A) All students named George smile. B) No student smiles, except George. C) Every student except George smiles. D) Only students named George smile. **Correct Answer: C**

47. Which type of inference is triggered by the arrival of new data in working memory? A) Backward Chaining B) Goal-Directed Inference C) Forward Chaining D) Hypothesis-Driven Inference **Correct Answer: C**

48. Which type of inference starts by trying to prove a particular goal or hypothesis? A) Forward Chaining B) Data-Directed Inference C) Backward Chaining D) Fact-Driven Inference **Correct Answer: C**

49. In the FOL expression $\text{Sqrt}(4)$, what does Sqrt represent? A) Predicate B) Constant C) Variable D) Function **Correct Answer: D**

50. What is the primary limitation of Propositional Logic that FOL addresses? A) Inability to represent facts. B) Inability to use logical connectives (AND, OR, NOT). C) Inability to represent objects, properties, relations, and use quantifiers. D) Inability to perform logical inference. **Correct Answer: C**

51. If we know runny_nose is true, and have the rule $\text{IF runny_nose THEN nasal_congestion}$, which inference method directly uses this to conclude nasal_congestion ? A) Backward Chaining B) Forward Chaining C) Resolution D) Abduction **Correct Answer: B**

52. If our goal is to find out if $\text{diagnose}(\text{influenza})$ is true, and we have the rule $\text{IF nasal_congestion AND virosis THEN diagnose}(\text{influenza})$, which inference method would use this rule to set nasal_congestion and virosis as subgoals? A) Forward Chaining B) Data-Directed Inference C) Backward Chaining D) Modus Ponens **Correct Answer: C**

53. The statement $\forall x \neg \text{talk}(x)$ is logically equivalent to: A) $\exists x \text{talk}(x)$ B) $\neg \exists x \text{talk}(x)$ C) $\forall x \text{talk}(x)$ D) $\neg \forall x \text{talk}(x)$ **Correct Answer: B**
54. The statement $\neg \forall x \neg \text{plays}(x, \text{Cricket})$ is logically equivalent to: A) $\forall x \text{plays}(x, \text{Cricket})$ B) $\exists x \text{plays}(x, \text{Cricket})$ C) $\neg \exists x \text{plays}(x, \text{Cricket})$ D) $\forall x \neg \text{plays}(x, \text{Cricket})$ **Correct Answer: B**
55. What component of an expert system is responsible for storing temporary, case-specific information provided by the user? A) Knowledge Base B) Inference Engine C) User Interface D) Working Memory (or Case-Specific Facts) **Correct Answer: D**
56. What does the equality symbol $==$ typically represent in FOL discussions? A) Assignment B) Implication C) Equivalence or Identity D) Conjunction **Correct Answer: C**
57. In the context of the medical examples, what acts as the 'domain knowledge'? A) The initial patient symptoms (working memory). B) The set of IF-THEN rules (R1-R6). C) The final diagnosis. D) The inference engine's algorithm. **Correct Answer: B**
58. Which quantifier is associated with the connector 'AND' ($\wedge$) in typical translations? A) $\forall$ (Universal) B) $\exists$ (Existential) C) $\neg$ (Negation) D) $\rightarrow$ (Implication) **Correct Answer: B**
59. Which quantifier is associated with the connector 'IF...THEN' ($\rightarrow$) in typical translations? A) $\forall$ (Universal) B) $\exists$ (Existential) C) $\wedge$ (Conjunction) D) $\vee$ (Disjunction) **Correct Answer: A**
60. If an expert system needs to find out a specific piece of information (e.g., patient's temperature) and searches for rules that conclude this information, what inference strategy is it likely using? A) Forward Chaining B) Backward Chaining C) Data Mining D) Pattern Matching **Correct Answer: B**

MCQs from: Integration of Semantic Networks, Frames, and ontologies in ES.pdf (pdf2_integration.txt)

Instructions: Choose the best answer for each question.

1. What are the core components of an expert system mentioned in the recap? A) User Interface and Database B) Knowledge Base and Inference Engine C) Algorithm and Data Structure D) Input Sensor and Output Actuator **Correct Answer: B**
2. Why is selecting an appropriate Knowledge Representation (KR) method critical in expert systems? A) It determines the programming language used. B) It dictates the user interface design. C) It directly impacts inference engine function and problem-solving efficiency. D) It defines the hardware requirements. **Correct Answer: C**
3. Which KR techniques are mentioned as common in the document? A) Decision Trees, Neural Networks, Rule-Based Systems B) Semantic Networks, Frames, Ontologies C) Flowcharts, State Machines, Petri Nets D) Databases, Spreadsheets, Text Files **Correct Answer: B**
4. What is the primary role of the Inference Engine in an expert system? A) Storing domain-specific information. B) Providing explanations for conclusions. C) Interacting directly with the user. D) Drawing conclusions based on the knowledge base (reasoning core). **Correct Answer: D**
5. Which of the following is NOT listed as a key responsibility of the inference engine cycle? A) Match rules with facts. B) Select/Resolve Conflicts among matching rules. C) Act/Fire the selected rule. D) Acquire new knowledge from external sources. **Correct Answer: D**
6. In semantic networks, what do nodes typically represent? A) Relationships B) Rules C) Concepts or entities D) Inference steps **Correct Answer: C**
7. In semantic networks, what do edges typically represent? A) Concepts or entities B) Rules C) Relationships between concepts/entities D) Facts **Correct Answer: C**
8. What type of rule in semantic networks handles property propagation via 'is-a' links? A) Compositional B) Constraint C) Inheritance D) Associative **Correct Answer: C**

9. The example "Dog is-a Mammal → Dog has-a Heart" illustrates which type of rule?
A) Classification B) Inheritance C) Equivalence D) Default/Defeasible **Correct Answer: B**
Answer: B
10. What type of rule allows inference via 'has-a' relationships? A) Inheritance B) Compositional C) Constraint D) Equivalence **Correct Answer: B**
11. The rule "Vehicle with Wings → Must be Aircraft" is an example of which rule type?
A) Associative B) Classification C) Constraint D) Default/Defeasible **Correct Answer: C**
12. The rule "Has Feathers + Flies → is-a Bird" is an example of which rule type? A) Inheritance B) Compositional C) Classification D) Constraint **Correct Answer: C**
13. What type of rule handles common-sense logic with exceptions, like "Bird → can Fly; but Penguin → can't Fly"? A) Inheritance B) Default/Defeasible C) Constraint D) Associative **Correct Answer: B**
14. What is forward chaining described as in the context of semantic networks? A) Goal-driven inference B) Hypothesis-driven inference C) Data-driven inference D) User-driven inference **Correct Answer: C**
15. How does forward chaining operate in a semantic network? A) Starts with a goal and works backward. B) Starts with known facts and propagates inferences through relationships. C) Asks the user questions to guide the process. D) Randomly explores the network. **Correct Answer: B**
16. In the forward chaining example (Cat is-a Mammal, Mammal is-a Animal, Rule: If X is-an Animal, then X has-a Heart), what is the final inferred fact? A) Mammal has-a Heart B) Animal has-a Heart C) Cat is-an Animal D) Cat has-a Heart **Correct Answer: D**
Answer: D
17. In the second forward chaining example (Dog), what fact allows the inference "Dog is-a Animal"? A) Dog has-a Tail B) Dog is-a Mammal and Mammal is-a Animal (inheritance) C) Animal has-a Heart D) The rule "IF X is-a Animal THEN X has-a Heart" **Correct Answer: B**
18. What is the goal in the backward chaining example provided for the semantic network? A) Prove Dog has-a Heart B) Prove Dog is-a Animal C) Prove Dog is a Candidate for Surgery D) Prove Dog has-a Tail **Correct Answer: C**
19. In the backward chaining example, what are the two immediate subgoals required to prove "Dog is a Candidate for Surgery"? A) Dog is-a Mammal AND Dog is-a

Animal B) Dog has-a Heart AND Dog has-a Tail C) Dog is-a Animal AND Dog has-a Tail D) Dog is-a Mammal AND Dog has-a Heart **Correct Answer: B**

20. How is the subgoal "Dog has-a Heart" proven in the backward chaining example?
A) It is a given fact. B) By using the rule "IF X is-a Animal THEN X has-a Heart" and proving the sub-subgoal "Dog is-a Animal". C) By asking the user. D) It cannot be proven with the given information. **Correct Answer: B**
21. What is the first step in the Unified Step-by-Step Design Process for KR? A) Identify Key Concepts B) Define the Domain and Scope C) Identify Relationships D) Build the Class Hierarchy **Correct Answer: B**
22. What is the purpose of writing competency questions in Step 1 (Define Domain and Scope)? A) To test the final system. B) To clarify what decisions the system should make or support. C) To identify key concepts. D) To define relationships between concepts. **Correct Answer: B**
23. What is the main activity in Step 2 (Identify Key Concepts)? A) Defining the system boundaries. B) Capturing domain entities like classes, objects, roles, and processes. C) Defining relationships like 'is-a' and 'has-a'. D) Building the inheritance structure. **Correct Answer: B**
24. In Step 3 (Identify Relationships), what kind of relationships are defined? A) Only hierarchical ('is-a') relationships. B) Only compositional ('has-a') relationships. C) Formal semantic relationships (e.g., is-a, has-symptom, treated-by). D) Only user-system interaction relationships. **Correct Answer: C**
25. What is the main goal of Step 4 (Build the Class/Frame Hierarchy)? A) To define individual instances. B) To add attributes and properties. C) To organize concepts using inheritance structures (taxonomy). D) To define inference rules. **Correct Answer: C**
26. How does building a class hierarchy enable inferencing? A) By defining specific rules for each class. B) By allowing properties to be inherited down the hierarchy (e.g., Asthma inherits properties of RespiratoryDisease). C) By linking classes to external databases. D) By defining constraints on attributes. **Correct Answer: B**
27. What is the focus of Step 5 (Define Attributes / Slots / Properties)? A) Organizing classes hierarchically. B) Adding metadata (details like severity, age, symptoms) to classes or frames. C) Defining logical rules. D) Creating individual instances. **Correct Answer: B**
28. In Step 6 (Define Constraints and Rules), what might be added using Ontology tools like OWL/SWRL? A) Only class definitions. B) Only individual instances. C)

Cardinality constraints and inference rules. D) User interface elements. **Correct Answer: C**

29. What happens in Step 7 (Instantiate Individuals)? A) The class hierarchy is defined. B) Rules are written. C) Data-level examples or specific cases (like 'JohnDoe') are provided. D) The system scope is defined. **Correct Answer: C**
30. What is the purpose of Step 8 (Validate Against Competency Questions)? A) To define the initial scope. B) To identify key concepts. C) To simulate reasoning scenarios and check coverage and consistency. D) To create individual instances. **Correct Answer: C**
31. What tools might be used for validation in Step 8? A) Text editors and spreadsheets. B) UML diagramming tools. C) Ontology reasoners (HermiT), query languages (SPARQL), rule engines (CLIPS, Drools). D) Compilers and debuggers for general programming languages. **Correct Answer: C**
32. What is the goal of Step 9 (Iterate and Refine)? A) To define the initial domain. B) To create the first version of the knowledge base. C) To improve quality, modularity, and inference efficiency. D) To deploy the system to users. **Correct Answer: C**
33. In the Fitness Ontology example, what does the class 'WorkoutPlan' represent? A) A specific person using the system. B) A fitness goal like 'Weight Loss'. C) A structured set of exercises assigned based on goals. D) A category like 'Beginner' or 'Advanced'. **Correct Answer: C**
34. In the Fitness Ontology example, 'JohnDoe' is an instance (individual) of which class? A) WorkoutPlan B) Goal C) Exercise D) Person **Correct Answer: D**
35. Which relationship connects a 'Person' to their 'WorkoutPlan' in the Fitness Ontology example? A) hasGoal B) includesExercise C) assignedPlan D) hasFitnessLevel **Correct Answer: C**
36. Which relationship connects a 'WorkoutPlan' to the 'Exercises' it contains? A) hasGoal B) includesExercise C) assignedPlan D) hasIntensityLevel **Correct Answer: B**
37. According to Rule 1 in the Fitness Ontology, what triggers the assignment of Workout Plan A? A) The person being at an advanced fitness level. B) The person having the goal of building muscle. C) The person being older than 45. D) The workout plan including push-ups. **Correct Answer: B**
38. According to Rule 3 in the Fitness Ontology, what classifies a workout plan as high intensity? A) Being assigned to an advanced user. B) Having the goal of muscle

gain. C) Including push-ups as an exercise. D) Being named 'PlanA'. **Correct**

Answer: C

39. In the Fitness Ontology forward chaining example, what is the starting fact? A) assignedPlan(JohnDoe, PlanA) B) hasGoal(JohnDoe, BuildMuscle) C) includesExercise(PlanA, PushUp) D) hasIntensityLevel(PlanA, HighIntensity)

Correct Answer: B

40. What is the inferred fact after applying Rule 1 in the Fitness Ontology forward chaining example? A) hasGoal(JohnDoe, BuildMuscle) B) assignedPlan(JohnDoe, PlanA) C) includesExercise(PlanA, PushUp) D) JohnDoe is AdvancedLevel **Correct**

Answer: B

41. In the Fitness Ontology backward chaining example, what is the initial goal? A) Is JohnDoe assigned PlanA? B) Does JohnDoe have the goal BuildMuscle? C) Is PlanA high intensity? D) Does PlanA include PushUp? **Correct Answer: C** (Goal: hasIntensityLevel(PlanA, HighIntensity))

42. Which rule is used in the backward chaining example to potentially prove the goal "Is PlanA high intensity?" A) Rule 1 B) Rule 2 C) Rule 3 D) Rule 4 **Correct Answer: C**

43. What sub-goal (question) arises from applying Rule 3 in the backward chaining example? A) Does PlanA have the goal BuildMuscle? B) Is JohnDoe assigned PlanA? C) Is JohnDoe AdvancedLevel? D) Does PlanA include PushUp? **Correct Answer: D**

44. What is the function of conflict resolution strategies (priority, specificity, recency) in an inference engine? A) To decide which facts are true. B) To choose which rule to fire when multiple rules match the current facts. C) To determine the initial goal for backward chaining. D) To structure the knowledge base. **Correct Answer: B**

45. Semantic networks primarily represent knowledge using: A) IF-THEN rules B) Nodes and labeled edges (graphs) C) Slots and facets within frames D) Axioms and logical formulas **Correct Answer: B**

46. Ontologies, compared to semantic networks, often provide: A) Less formal structure. B) More formal semantics and structure, often based on logic. C) Only visual representations. D) No support for inheritance. **Correct Answer: B**

47. The process of deriving "Cat has-a Heart" from "Cat is-a Mammal", "Mammal is-a Animal", and "Animal has-a Heart" relies heavily on: A) Default reasoning B) Associative rules C) Inheritance and transitivity D) Constraint checking **Correct**

Answer: C

48. What does 'saturation' mean in the context of forward chaining? A) The working memory is full. B) The goal has been reached. C) No new information can be inferred by applying rules to the current facts. D) The user has stopped the inference process. **Correct Answer: C**
49. Which KR step involves capturing domain entities like classes, objects, and processes, often through expert interviews? A) Step 1: Define Domain and Scope B) Step 2: Identify Key Concepts C) Step 3: Identify Relationships D) Step 4: Build Hierarchy **Correct Answer: B**
50. Representing relationships like 'is-a', 'has-symptom', 'treated-by' using slots with facets and procedural attachments is characteristic of which KR technique mentioned in Step 3? A) Semantic Networks B) Ontologies C) Frames D) Rule-Based Systems **Correct Answer: C**
51. Using tools like OWL (Web Ontology Language) and SWRL (Semantic Web Rule Language) is most associated with which KR technique in Step 6? A) Frames B) Semantic Networks C) Ontologies D) Propositional Logic **Correct Answer: C**
52. Adding specific examples like 'Pneumonia' as an instance of 'Disease' occurs in which step of the design process? A) Step 4: Build Hierarchy B) Step 5: Define Attributes C) Step 6: Define Rules D) Step 7: Instantiate Individuals **Correct Answer: D**
53. Using a reasoner like HermiT to find logical conflicts is part of which step? A) Step 2: Identify Concepts B) Step 5: Define Attributes C) Step 7: Instantiate Individuals D) Step 9: Iterate and Refine **Correct Answer: D**
54. The relationship 'hasFitnessLevel' in the Fitness Ontology connects which two types of concepts/classes? A) Person and Goal B) Person and FitnessLevel C) WorkoutPlan and Exercise D) WorkoutPlan and IntensityLevel **Correct Answer: B**
55. Rule 4 in the Fitness Ontology ("If a person is assigned Workout Plan A, then we can infer that their goal is to build muscle") is essentially the inverse of which other rule? A) Rule 1 B) Rule 2 C) Rule 3 D) Rule 5 **Correct Answer: A**
56. What kind of reasoning is demonstrated by Rule 5 ("If a person is older than 45 years, then their fitness level should be set to beginner")? A) Inheritance B) Classification based on attributes C) Compositional reasoning D) Backward chaining **Correct Answer: B**
57. The unified design process emphasizes moving from: A) Code to documentation B) Specific instances to general rules C) Domain understanding to formal knowledge representation D) User interface to inference engine **Correct Answer: C**

58. Associative rules in semantic networks represent: A) Strict inheritance B) Part-whole relationships C) Indirect relationships or co-occurrence D) Logical constraints **Correct Answer: C**
59. The example "Doctor uses Stethoscope → Related to Heart Diagnosis" illustrates which type of rule? A) Inheritance B) Compositional C) Constraint D) Associative **Correct Answer: D**
60. When backward chaining attempts to prove a subgoal (e.g., "Dog has-a Heart") and finds a rule whose conclusion matches (e.g., "IF X is-a Animal THEN X has-a Heart"), what becomes the new subgoal? A) The conclusion of the rule (X has-a Heart) B) The premise/condition of the rule (X is-a Animal) C) The original top-level goal D) A fact from the working memory **Correct Answer: B**

MCQs from: Semantic Networks _ Lecture 9.pdf (pdf3_semantic_networks.txt)

Instructions: Choose the best answer for each question.

1. What is a semantic network? A) A type of neural network. B) A database schema. C) A graphical representation of knowledge with concepts (nodes) connected by relationships (edges). D) A programming language for AI. **Correct Answer: C**
2. In a semantic network, what do nodes represent? A) Relationships B) Rules C) Concepts or entities D) Actions **Correct Answer: C**
3. In a semantic network, what do edges (links) represent? A) Concepts or entities B) The strength of a connection C) Relationships between nodes D) Data types **Correct Answer: C**
4. According to the text, what is a primary benefit of using semantic networks in AI? A) They guarantee optimal decision-making. B) They facilitate data organization, interpretation, and utilization for decision-making. C) They are the only way to represent hierarchical knowledge. D) They automatically learn from unstructured data. **Correct Answer: B**
5. Which relationship type signifies that one concept belongs to the category of another (e.g., "Dog" to "Animal")? A) has-a B) part-of C) is-a D) associated-with **Correct Answer: C**

6. Which of the following is NOT listed as a key component of semantic networks? A) Nodes B) Edges C) Algorithms D) Labels **Correct Answer: C**
7. What do labels on edges typically specify? A) The size of the node. B) The nature or type of the relationship between nodes. C) The certainty of the relationship. D) The computational cost of traversing the edge. **Correct Answer: B**
8. The relationship "Wheel" is part of "Car" is an example of which label type? A) Is-A B) Part-Of C) Has-A D) Instance-Of **Correct Answer: B**
9. The relationship "Car" has "Wheels" is an example of which label type? A) Is-A B) Part-Of C) Has-A D) Instance-Of **Correct Answer: C**
10. The relationship "Rover" is an instance of "Dog" is an example of which label type? A) Is-A B) Part-Of C) Has-A D) Instance-Of **Correct Answer: D**
11. What role do attributes or properties play in semantic networks? A) They define the primary relationships between nodes. B) They provide additional descriptive information about the concepts represented by nodes. C) They represent actions the network can perform. D) They are always modeled as edges. **Correct Answer: B**
12. What is inheritance in the context of semantic networks? A) The process of creating new nodes. B) The ability for child nodes to receive properties or relationships from parent nodes. C) The measure of distance between two nodes. D) The labeling of edges. **Correct Answer: B**
13. If "Animal" has the property "can move", and "Dog" is-a "Animal", what can be inferred via inheritance? A) "Animal" can move. B) "Dog" can move. C) "Move" is an instance of "Animal". D) All nodes can move. **Correct Answer: B**
14. What does "semantic distance" typically refer to in a semantic network? A) The physical distance between nodes on a diagram. B) The number of attributes a node has. C) How closely related two concepts are, often measured by the path length between them. D) The time it takes to process a relationship. **Correct Answer: C**
15. Which type of semantic network primarily represents hierarchical relationships (subtype/supertype)? A) Assertional Networks B) Implicational Networks C) Definitional Networks D) Executable Networks **Correct Answer: C**
16. What is the main focus of Assertional Networks? A) Defining general categories. B) Asserting facts about specific instances or individuals. C) Representing logical implications. D) Representing procedural knowledge. **Correct Answer: B**

17. The statement "Rex is a Dog" and "Rex has Brown Fur" would typically be found in which type of network? A) Definitional Network B) Assertional Network C) Implicational Network D) Learning Network **Correct Answer: B**
18. Which type of network focuses on representing logical implications to infer new knowledge? A) Assertional Networks B) Implicational Networks C) Executable Networks D) Definitional Networks **Correct Answer: B**
19. Inferring "Rex is a Mammal" from "All Dogs are Mammals" and "Rex is a Dog" is characteristic of which network type? A) Assertional Network B) Definitional Network C) Implicational Network D) Executable Network **Correct Answer: C**
20. Which type of network is designed to represent procedural knowledge, like steps in a process? A) Implicational Networks B) Learning Networks C) Hybrid Networks D) Executable Networks **Correct Answer: D**
21. Representing the steps "Add Water to Pot" followed by "Boil Water" is an example use of which network type? A) Definitional Network B) Assertional Network C) Executable Network D) Implicational Network **Correct Answer: C**
22. Which type of network is dynamic and updates its structure based on new data or experiences? A) Definitional Networks B) Assertional Networks C) Learning Networks D) Executable Networks **Correct Answer: C**
23. Which type of network combines elements from two or more other types? A) Definitional Networks B) Assertional Networks C) Implicational Networks D) Hybrid Networks **Correct Answer: D**
24. In the "Healthy Lifestyle" example, what is the inferred consequence of "sleeps early"? A) Misses breakfast B) Feels tired by noon C) Performs well in work or study D) Wakes up late **Correct Answer: C**
25. In the "Healthy Lifestyle" example, what is the direct consequence of "wakes up late"? A) Has time for exercise B) Feels energetic C) Misses breakfast D) Performs well in work **Correct Answer: C**
26. According to the "Healthy Lifestyle" network, can a person who sleeps late feel energetic in the morning? Why? A) Yes, if they exercise. B) Yes, there is no link preventing it. C) No, because sleeping late leads to waking up late, missing breakfast, and feeling tired. D) No, because sleeping late directly causes tiredness. **Correct Answer: C**
27. How do semantic networks aid Natural Language Processing (NLP)? A) By defining grammar rules. B) By helping computers understand word meanings and

relationships (disambiguation). C) By translating languages directly. D) By generating natural language text. **Correct Answer: B**

28. What is word sense disambiguation in NLP, aided by semantic networks? A) Correcting spelling errors. B) Identifying the correct meaning of a word based on context and related words. C) Translating words to different languages. D) Summarizing text. **Correct Answer: B**
29. How are semantic networks used in Expert Systems? A) To provide the user interface. B) To store the inference engine logic. C) To model domain-specific knowledge (concepts and relationships) for reasoning. D) To execute external programs. **Correct Answer: C**
30. What is an ontology, often built using semantic networks? A) A type of expert system. B) A machine learning algorithm. C) A structured map of concepts and relationships in a specific domain. D) A natural language processing technique. **Correct Answer: C**
31. How do semantic networks enhance Information Retrieval (search engines)? A) By indexing web pages faster. B) By understanding the context and conceptual relationships between search terms and documents. C) By storing user search history. D) By displaying search results graphically. **Correct Answer: B**
32. How can semantic networks be used in Machine Learning? A) As a replacement for training data. B) To enhance model interpretability by representing relationships between learned concepts. C) To directly classify data points. D) To optimize the learning rate. **Correct Answer: B**
33. Which of the following is listed as an advantage of semantic networks? A) Guaranteed scalability B) Intuitive representation C) Perfect expressiveness for all knowledge types D) Ease of representing temporal relationships **Correct Answer: B**
34. Which of the following is listed as a disadvantage of semantic networks? A) Rigid structure B) Poor contextual understanding C) Scalability issues in large systems D) Inefficient inference **Correct Answer: C**
35. What potential issue arises with maintaining semantic networks over time? A) They become too simple. B) Relationships and concepts may evolve, making updates challenging. C) They lose their graphical nature. D) They cannot be expanded. **Correct Answer: B**
36. What limitation regarding expressiveness is mentioned for semantic networks? A) Cannot represent hierarchical relationships. B) May struggle with complex/

nuanced knowledge like ambiguity or probabilities. C) Cannot represent simple facts. D) Cannot use labels on edges. **Correct Answer: B**

37. Semantic networks may have difficulty representing which type of information? A) Hierarchical classifications B) Part-whole relationships C) Static facts D) Time-dependent information or dynamic changes **Correct Answer: D**
38. The ability to easily expand and adapt semantic networks relates to which advantage? A) Intuitive Representation B) Flexible Structure C) Efficient Inference D) Interoperability **Correct Answer: B**
39. The ability of semantic networks to facilitate knowledge sharing across systems relates to which advantage? A) Contextual Understanding B) Flexible Structure C) Interoperability D) Intuitive Representation **Correct Answer: C**
40. If a network shows "Penguin is-a Bird" and "Bird has-property Flies", but also "Penguin has-property Not-Flies", this illustrates a potential challenge related to: A) Scalability B) Inheritance with exceptions (often handled by default reasoning, related to expressiveness limits) C) Information Retrieval D) Network visualization **Correct Answer: B**
41. Definitional networks are also known as: A) Assertional networks B) Implicational networks C) Taxonomic or subtype networks D) Executable networks **Correct Answer: C**
42. Assertional networks are particularly useful in expert systems for representing: A) General rules B) Hierarchies C) Facts about individual patients or cases D) Procedural steps **Correct Answer: C**
43. Implicational networks are fundamental in: A) Storing instance data B) Visualizing hierarchies C) Rule-based systems and automated reasoning D) Representing procedural tasks **Correct Answer: C**
44. Executable networks are often used in AI systems that perform tasks requiring: A) Classification B) Diagnosis C) A sequence of actions (procedural knowledge) D) Natural language understanding **Correct Answer: C**
45. Learning networks are characterized by their ability to: A) Remain static once created. B) Evolve and update based on new information or experiences. C) Represent only procedural knowledge. D) Focus solely on hierarchical definitions. **Correct Answer: B**
46. A network representing both the general concept of "Car" (definitional) and specific instances like "MyTesla" (assertional) would be considered a: A) Learning

Network B) Executable Network C) Hybrid Network D) Implicational Network

Correct Answer: C

47. The measure of how closely related two nodes are in a network is called: A) Node degree B) Edge weight C) Semantic distance D) Network diameter **Correct Answer: C**
48. In NLP, semantic networks help overcome the problem where words have multiple meanings depending on context. This problem is called: A) Syntax error B) Ambiguity (requiring disambiguation) C) Stemming D) Tokenization **Correct Answer: B**
49. Using semantic networks to represent relationships between learned concepts in ML aims to improve: A) Training speed B) Model accuracy C) Model interpretability D) Data storage efficiency **Correct Answer: C**
50. The example "Dog" linked to "Animal" via "is a" primarily demonstrates which key component? A) Attributes B) Inheritance C) Nodes and a Labeled Edge D) Semantic Distance **Correct Answer: C**
51. Which network type would be most suitable for representing the knowledge: "If temperature is below freezing AND precipitation is falling, THEN road conditions are icy"? A) Definitional Network B) Assertional Network C) Implicational Network D) Executable Network **Correct Answer: C**
52. Which network type would be most suitable for representing the knowledge: "My specific car, VIN 123, is blue"? A) Definitional Network B) Assertional Network C) Implicational Network D) Learning Network **Correct Answer: B**
53. Which network type would be most suitable for representing the knowledge: "To make tea: Boil water, Add tea bag, Steep for 3 minutes"? A) Definitional Network B) Assertional Network C) Implicational Network D) Executable Network **Correct Answer: D**
54. Which network type would be most suitable for representing the knowledge: "A poodle is a type of dog, which is a type of mammal, which is a type of animal"? A) Definitional Network B) Assertional Network C) Executable Network D) Learning Network **Correct Answer: A**
55. A potential disadvantage where semantic networks might struggle to represent that "Tweety is a bird" implies "Tweety usually flies", but knowing "Tweety is a penguin" overrides this, relates to: A) Scalability B) Limited expressiveness (handling exceptions/defaults) C) Interoperability D) Visualization **Correct Answer: B**

56. The ease with which humans can often grasp the structure of a semantic network relates to which advantage? A) Flexible Structure B) Interoperability C) Efficient Inference D) Intuitive Representation **Correct Answer: D**
57. The ability to deduce "Rover can bark" if "Dog can bark" and "Rover is-a Dog" is an example of: A) Semantic Distance Calculation B) Inference (often using inheritance) C) Network Learning D) Ambiguity Resolution **Correct Answer: B**
58. According to the text, attributes (like 'color') are sometimes modeled as separate nodes, especially if: A) They are unique to one entity. B) They are numeric. C) They are shared across entities or have their own properties. D) They represent actions. **Correct Answer: C**
59. The primary goal of using semantic networks in AI, as stated early in the text, is to: A) Create visually appealing diagrams. B) Replace traditional databases. C) Facilitate data organization for machine processing and decision-making. D) Generate human-like conversation. **Correct Answer: C**
60. Which application involves using semantic networks to create structured maps of concepts and relationships within a specific field of knowledge? A) Natural Language Processing B) Expert Systems C) Ontology Development D) Machine Learning **Correct Answer: C**

MCQs from: lecture 12_Frames in ES and KR practice question.pdf (pdf4_frames_practice.txt)

Instructions: Choose the best answer for each question.

1. What is the core type of inference in frame systems where a subclass or instance receives slot values from its parent? A) Default Value Inference B) Procedural Attachment Inference C) Inheritance Inference D) Rule-Based Inference **Correct Answer: C**
2. In inheritance inference, what happens if a slot value is explicitly defined in a child frame? A) The parent value is used. B) An error occurs. C) The child's value overrides the inherited parent value. D) The default value is used. **Correct Answer: C**

3. Consider the frames: `Animal` (`hasLegs: 4`) and `Dog` (`inherits from Animal`). If `Dog` frame doesn't define `hasLegs`, what value does `Dog.hasLegs` inherit? A) 2 B) 4 C) undefined D) default **Correct Answer: B**
4. When is Default Value Inference used in frames? A) When a slot value is inherited from a parent. B) When a specific value is not defined for a slot in an instance, and the slot has a default specified. C) When a procedural attachment is triggered. D) Only for slots that represent numbers. **Correct Answer: B**
5. Consider the frame `Course` (`hasFinalExam: default true`). If an instance `Course101` does not define `hasFinalExam`, what is the value of `Course101.hasFinalExam`? A) false B) undefined C) true (from default) D) Inherited from a parent frame **Correct Answer: C**
6. What allows a frame to run code when certain conditions are met, like accessing or modifying a slot? A) Inheritance B) Default Values C) Slot Constraints D) Procedural Attachments **Correct Answer: D**
7. Which type of procedural attachment is triggered when a slot is accessed and its value is missing? A) IF-ADDED B) DEFAULT C) IF-NEEDED D) ON-ACCESS **Correct Answer: C**
8. What is the primary purpose of an IF-NEEDED procedure? A) To validate a new value added to a slot. B) To provide a default value. C) To compute and return the slot's value dynamically when it's accessed and missing. D) To log access to the slot. **Correct Answer: C**
9. In the `HospitalPatient` example, accessing the `RoomNumber` slot triggers `getAvailableRoom()` because `RoomNumber` uses which type of attachment? A) IF-ADDED B) DEFAULT C) IF-NEEDED D) INHERITED **Correct Answer: C**
10. What is a key benefit of using IF-NEEDED procedures, as mentioned in the text? A) It simplifies the frame structure. B) It ensures values are always pre-computed. C) It supports lazy evaluation (computes only when needed). D) It restricts access to slots. **Correct Answer: C**
11. Which type of procedural attachment is triggered when a new value is written into a slot? A) IF-NEEDED B) IF-ADDED C) DEFAULT D) ON-WRITE **Correct Answer: B**
12. What is a common purpose for an IF-ADDED procedure? A) To compute a value when read. B) To trigger a function for validation or logging when a value is added. C) To inherit a value from a parent. D) To provide a fallback value. **Correct Answer: B**

13. In the `Student / Course` example, adding `Mona` to the `EnrollStudent` slot of `CS301` triggers `validatePrerequisites(Mona)`. This is an example of: A) IF-NEEDED B) DEFAULT C) Inheritance D) IF-ADDED **Correct Answer: D**
14. What happens if a slot has a DEFAULT procedural attachment? A) It runs code every time the slot is accessed. B) It provides a fallback value if no specific value is given, unless overridden. C) It triggers validation when a value is added. D) It computes the value dynamically when needed. **Correct Answer: B**
15. In the `Course` example with `FinalExamDuration` having a default of 90 mins, what is the duration for `IS201` where it's not provided? A) 120 mins B) 90 mins (default used) C) undefined D) Inherited value **Correct Answer: B**
16. In the `Course` example, what is the `FinalExamDuration` for `ML402` which explicitly defines it as 120 mins? A) 90 mins (default) B) 120 mins (explicit value overrides default) C) undefined D) Calculated by a procedure **Correct Answer: B**
17. In the TV manufacturing example, the `TV` frame inherits from which frame? A) `SmartTV` B) `Component` C) `ElectronicDevice` D) `FactoryAssemblyLine` **Correct Answer: C**
18. The `SmartTV` frame inherits from which frame? A) `ElectronicDevice` B) `TV` C) `Component` D) `FactoryAssemblyLine` **Correct Answer: B**
19. What is the default `warrantyPeriod` for an `ElectronicDevice`? A) 6 months B) 12 months C) 24 months D) Not specified **Correct Answer: B**
20. If `TVModel_X100` is an instance of `SmartTV` and doesn't define `warrantyPeriod`, what value does it take? A) The value from `TV`. B) The default value (12 months) from `ElectronicDevice` through inheritance. C) It remains undefined. D) It triggers an IF-NEEDED procedure. **Correct Answer: B**
21. The `getPrice` slot in the `TV` frame uses which type of procedural attachment? A) IF-ADDED B) DEFAULT C) IF-NEEDED D) None **Correct Answer: C**
22. When is the `calculatePrice()` procedure executed for `TVModel_X100`? A) When the frame is created. B) When the `getPrice` slot is accessed and no value is stored. C) When a component is added. D) Periodically. **Correct Answer: B**
23. The execution of `calculatePrice()` only when the `getPrice` slot is accessed demonstrates what concept? A) Eager evaluation B) Default reasoning C) Lazy inference/evaluation D) Conflict resolution **Correct Answer: C**

24. The `addComponent` slot in the `Component` frame uses which type of procedural attachment? A) IF-NEEDED B) IF-ADDED C) DEFAULT D) None **Correct Answer: B**
25. What does the `logComponentAddition()` procedure do when triggered in the TV example? A) Calculates the price of the component. B) Checks if the component is compatible. C) Logs the component's supplier and addition date. D) Assigns the component to a TV model. **Correct Answer: C**
26. In the artwork example, the `Painting` frame inherits from which frame? A) Sculpture B) Artist C) Exhibition D) Artwork **Correct Answer: D**
27. What is the default value for the `year` slot in the `Artwork` frame? A) 2024 B) 0 C) unknown D) Current year **Correct Answer: C**
28. What is the default value for the `installationRequired` slot in the `Sculpture` frame? A) false B) true C) unknown D) Depends on material **Correct Answer: B**
29. The `getEstimatedValue` slot in the `Artwork` frame uses which procedural attachment? A) IF-ADDED B) DEFAULT C) IF-NEEDED D) LOG-ACCESS **Correct Answer: C**
30. The `addArtwork` slot in the `Artist` frame uses which procedural attachment? A) IF-NEEDED B) IF-ADDED C) DEFAULT D) VALIDATE **Correct Answer: B**
31. What does the `estimateValue()` procedure likely base its calculation on? A) Only the artist's name. B) Style, age, and medium. C) Exhibition history. D) Weight and dimensions. **Correct Answer: B**
32. What does the `logCreation()` procedure do when an artwork is added to an artist's portfolio? A) Estimates the artwork's value. B) Records the event (artwork addition). C) Checks if the artwork already exists. D) Assigns the artwork to an exhibition. **Correct Answer: B**
33. The `addArtwork` slot in the `Exhibition` frame uses which procedural attachment? A) IF-NEEDED B) IF-ADDED C) DEFAULT D) ESTIMATE **Correct Answer: B**
34. What information does the `logDisplayInfo()` procedure store? A) Artist's biography. B) Artwork creation details. C) Artwork exhibit history and curatorial notes. D) Estimated value changes. **Correct Answer: C**
35. In the artwork query example (

MCQs from: lecture 10 ontologies and frames.pdf (pdf5_ontologies_frames.txt)

Instructions: Choose the best answer for each question.

1. What is an ontology in the context of knowledge representation? A) A type of database schema. B) A graphical user interface. C) A formal, explicit specification of a shared conceptualization. D) A machine learning algorithm. **Correct Answer: C**
2. What does an ontology typically define? A) Programming functions and variables. B) User interaction flows. C) Classes, instances, relations, and axioms within a domain. D) Hardware specifications. **Correct Answer: C**
3. In ontology terms, what are 'Classes'? A) Specific objects or occurrences. B) Rules or constraints. C) Categories or types of objects in the domain. D) Connections between objects. **Correct Answer: C**
4. In ontology terms, what are 'Instances' (Individuals)? A) Categories or types of objects. B) Rules or constraints. C) Specific objects or occurrences within classes. D) Connections between objects. **Correct Answer: C**
5. What are 'Relations' (Properties) in an ontology? A) Categories or types of objects. B) Rules or constraints. C) Specific objects or occurrences. D) Connections between objects or individuals (e.g., "is a", "part of"). **Correct Answer: D**
6. What are 'Axioms' in an ontology? A) Categories or types of objects. B) Rules or constraints that define logical relationships and behaviors. C) Specific objects or occurrences. D) Connections between objects. **Correct Answer: B**
7. What is the first step in the Ontology Development Process described? A) Define Classes and Instances B) Define Relationships C) Domain Analysis D) Add Axioms and Constraints **Correct Answer: C**
8. What is involved in the 'Define Classes and Instances' step of ontology development? A) Understanding the domain. B) Specifying relationships. C) Creating a hierarchy of classes and defining specific instances. D) Defining logical rules. **Correct Answer: C**
9. What is the purpose of the 'Evaluation and Refinement' step in ontology development? A) To define the initial classes. B) To add logical rules. C) To continuously check the ontology for correctness, completeness, and usability. D) To perform domain analysis. **Correct Answer: C**

10. Why are ontologies important for Standardization and Interoperability? A) They make systems run faster. B) They provide a shared understanding allowing different systems to interpret information consistently. C) They automatically generate code. D) They define user interface standards. **Correct Answer: B**
11. How do ontologies support Reasoning and Inference? A) By storing large amounts of raw data. B) By enabling systems to derive new knowledge from existing facts and axioms. C) By providing graphical visualization tools. D) By defining programming logic. **Correct Answer: B**
12. How do ontologies improve Semantic Search? A) By indexing keywords faster. B) By providing context and understanding meaning, leading to more accurate results. C) By storing search history. D) By limiting search results to specific websites. **Correct Answer: B**
13. Which application area uses ontologies as its core to enable intelligent data integration and search? A) Robotics B) E-commerce C) Semantic Web D) Healthcare **Correct Answer: C**
14. How are ontologies used in Natural Language Processing (NLP)? A) To generate human-like speech. B) To correct grammatical errors. C) To help systems understand language by providing formal representations of concepts and relations. D) To translate text between languages. **Correct Answer: C**
15. In healthcare, ontologies like SNOMED CT are used primarily for: A) Billing patients. B) Scheduling appointments. C) Standardizing medical terminology for interoperability and research. D) Designing hospital layouts. **Correct Answer: C**
16. How are ontologies applied in E-commerce? A) To process payments. B) To categorize products for efficient search and recommendation. C) To manage customer accounts. D) To design website advertisements. **Correct Answer: B**
17. Which of the following is listed as an advantage of ontologies? A) Simplicity of creation B) Low initial cost C) Improved Knowledge Sharing D) Limited expressiveness **Correct Answer: C**
18. Which of the following is listed as a disadvantage of ontologies? A) Poor reasoning capabilities B) Lack of standardization C) Complexity and high cost of creation D) Inability to scale **Correct Answer: C**
19. What challenge is associated with maintaining consistency in large ontologies? A) They become too simple over time. B) Integrating new concepts without creating contradictions can be difficult. C) Performance improves significantly. D) They require less domain knowledge as they grow. **Correct Answer: B**

20. What is meant by the 'Limited Expressiveness' disadvantage of ontologies? A) They cannot represent any relationships. B) They struggle to represent complex or tacit (not easily formalized) knowledge. C) They can only be written in one language. D) They cannot be visualized. **Correct Answer: B**
21. According to the comparison table, how do Frames primarily represent knowledge? A) As a graph of nodes and edges. B) As a structured representation with slots and values. C) As formal logic axioms. D) As unstructured text. **Correct Answer: B**
22. According to the comparison table, what is the primary purpose of Semantic Networks? A) Representing objects with detailed attributes. B) Formal knowledge representation for reasoning. C) Representing relationships between concepts. D) Storing procedural knowledge. **Correct Answer: C**
23. Which knowledge representation method is described as 'Highly expressive, supports complex reasoning and constraints' in the table? A) Semantic Networks B) Frames C) Ontologies D) Rule-Based Systems **Correct Answer: C**
24. When is it most appropriate to use Frames, according to the comparison table? A) When representing simple concept mappings. B) When needing formalized knowledge with complex reasoning for domains like healthcare. C) When representing objects with attributes and inheritance (e.g., expert systems). D) When dealing primarily with graph structures. **Correct Answer: C**
25. When is it most appropriate to use Ontologies, according to the comparison table? A) For simple concept mapping. B) When representing objects with basic attributes. C) For early AI experiments. D) When needing formalized, structured knowledge with reasoning for complex domains. **Correct Answer: D**
26. What are Frames defined as in the text? A) A type of semantic network node. B) Data structures representing stereotyped situations or objects in a structured form. C) Logical axioms. D) A programming paradigm. **Correct Answer: B**
27. What are the components of a typical Frame? A) Nodes, Edges, Labels B) Classes, Instances, Relations, Axioms C) Slots, Fillers, Defaults, Inheritance mechanism D) Inputs, Outputs, Hidden Layers **Correct Answer: C**
28. In a Frame, what do 'Slots' represent? A) Specific values or data. B) Default assumptions. C) The attributes or properties of the object or concept. D) Links to parent frames. **Correct Answer: C**
29. In a Frame, what are 'Fillers'? A) The attributes or properties. B) Default assumptions. C) The specific values or data that populate the slots. D) Procedural attachments. **Correct Answer: C**

30. What is the role of 'Defaults' in a Frame? A) They represent the main attributes. B) They provide values or assumptions for slots that can be overridden. C) They define the relationships between frames. D) They are the specific data filling the slots.
Correct Answer: B
31. What does 'Inheritance' allow in Frame systems? A) Frames to execute code. B) Slots to have default values. C) Frames to inherit properties from parent frames in a hierarchy. D) Frames to link to external databases. **Correct Answer: C**
32. What is the first step described in 'How Frame Inheritance Works'? A) Create Child Frame B) Define Parent Frame C) Use Inherited Attributes D) Override or Extend
Correct Answer: B
33. What happens when a child frame is created in frame inheritance? A) It defines all slots from scratch. B) It automatically inherits slots and values from the parent. C) It loses connection to the parent frame. D) It can only inherit default values. **Correct Answer: B**
34. Can a child frame modify or add to the attributes inherited from a parent frame? A) No, inherited attributes are fixed. B) Yes, it can override inherited values and add its own unique slots. C) Only if the parent frame allows it. D) Only default values can be overridden. **Correct Answer: B**
35. In the Car example, the 'Car' frame inherits from which parent frame? A) Ford B) Vehicle C) Engine D) Transmission **Correct Answer: B**
36. In the Car example, 'Number of Doors' is what kind of slot in the 'Car' frame? A) An inherited slot from 'Vehicle'. B) An overridden slot from 'Vehicle'. C) An additional slot specific to 'Car'. D) A default slot. **Correct Answer: C**
37. In the Car example summary, the 'Year' slot in the 'Car' frame has the value 2023. This is an example of: A) Inheriting the value from 'Vehicle'. B) Using a default value. C) Overriding the inherited value (assuming 'Vehicle' might have a different or no year). D) An additional slot. **Correct Answer: C**
38. In the Animal Classification example, which frame is the most general parent frame? A) Lion B) Mammal C) Animal D) Species **Correct Answer: C**
39. The 'Mammal' frame inherits attributes from which frame? A) Lion B) Animal C) Itself D) No parent frame **Correct Answer: B**
40. The 'Lion' frame inherits attributes directly from which frame? A) Animal B) Mammal C) Species D) Habitat **Correct Answer: B**

41. What is the inheritance relationship shown for the Lion example? A) Animal → Mammal → Lion B) Lion → Mammal → Animal C) Mammal → Animal → Lion D) Animal → Lion → Mammal **Correct Answer: B** (Arrow indicates inheritance direction: Child → Parent)
42. In the Fitness Class example, 'fitness class' is described as a specialization of: A) instructor B) group activity C) workout D) gym studio **Correct Answer: B**
43. In the Fitness Class example, 'yoga mat' has a superclass of: A) fitness class B) equipment C) foam D) floor-based exercises **Correct Answer: B**
44. In the Fitness Class example, 'cardio workout' is a specialization of: A) fitness class B) strength workout C) workout D) instructor **Correct Answer: C**
45. The frame description for 'Hotel Bed' likely inherits from a more general frame like: A) Room B) Furniture C) Building D) Mattress **Correct Answer: B** (Implied by context, though not explicitly stated as parent)
46. The frame description for 'Hotel Room' likely includes slots for components like: A) City, Country, Continent B) Check-in time, Check-out time, Price C) Bed, Bath, Telephone, TV D) Manager, Receptionist, Cleaner **Correct Answer: C**
47. How are frames used in Natural Language Processing (NLP), according to the text? A) To generate speech. B) To translate languages. C) To interpret context by extracting key details (e.g., for booking requests). D) To check spelling and grammar. **Correct Answer: C**
48. How are frames used in Expert Systems? A) To provide the user interface. B) To structure and represent domain-specific knowledge (e.g., diseases, symptoms). C) To perform complex calculations. D) To connect to the internet. **Correct Answer: B**
49. How are frames used in Robotics? A) To control motor movements directly. B) To provide power to the robot. C) To represent objects and their characteristics for environmental understanding. D) To design the robot's physical appearance. **Correct Answer: C**
50. What is a key difference in structure between Semantic Networks and Frames? A) Semantic networks use slots, frames use nodes. B) Frames are hierarchical with slots/values, semantic networks are graphs of nodes/edges. C) Semantic networks support inheritance, frames do not. D) Frames are less expressive than semantic networks. **Correct Answer: B**
51. What is a key difference in purpose between Frames and Ontologies? A) Frames represent relationships, ontologies represent objects. B) Ontologies are for simple

mappings, frames are for complex reasoning. C) Frames focus on representing objects with attributes, ontologies focus on formal knowledge representation for reasoning. D) Ontologies do not support inheritance. **Correct Answer: C**

52. Which representation is generally considered the most formally structured and suitable for complex logical reasoning? A) Semantic Networks B) Frames C) Ontologies D) Flowcharts **Correct Answer: C**
53. If you need to represent a 'meeting' event with attributes like 'time', 'location', 'attendees', and 'agenda', which KR method would be most suitable according to its definition? A) Semantic Network B) Frame C) Ontology (could be used, but Frame fits the 'stereotyped situation' definition well) D) Rule-based system **Correct Answer: B**
54. The ability of a 'Car' frame to inherit 'Engine Type' from a 'Vehicle' frame demonstrates which feature of frames? A) Slots B) Fillers C) Defaults D) Inheritance **Correct Answer: D**
55. Setting a default value for 'warrantyPeriod' in an 'ElectronicDevice' frame utilizes which frame component? A) Slot B) Filler C) Default D) Inheritance **Correct Answer: C**
56. The process of understanding the domain and identifying key concepts is crucial for which KR development process? A) Only Frame creation B) Only Semantic Network creation C) Only Ontology development D) Ontology development (explicitly mentioned), but generally applicable to others too. **Correct Answer: D** (Explicitly step 1 for Ontology, but fundamental to all KR)
57. Which KR method is explicitly mentioned as being core to the Semantic Web? A) Frames B) Semantic Networks C) Ontologies D) Expert Systems **Correct Answer: C**
58. The example inference "Every lion is a mammal" + "Simba is a lion" => "Simba is a mammal" highlights the reasoning capability supported by: A) Frame defaults B) Semantic network visualization C) Ontology axioms and relations D) Procedural attachments in frames **Correct Answer: C**
59. Representing that a 'yoga mat' is a type of 'equipment' with specific 'material' and 'length' fits well with which KR structure? A) Semantic Network (using 'is-a' and 'has-property' edges) B) Frame (using inheritance/superclass and slots like 'material', 'length') C) Ontology (using classes and properties) D) All of the above could potentially model this. **Correct Answer: D** (Though Frames are explicitly used in the example provided in the text)

60. The comparison table suggests that Semantic Networks are best suited for: A) Complex reasoning and constraints. B) Representing objects with detailed attributes. C) Formal knowledge management in healthcare. D) Simple concept mapping and representing basic relationships. **Correct Answer: D**